Session 3 | Linear Algebra I

Exercise sheet

Work through these in order: they start with simple vector arithmetic and build toward systems, inverses, and rank. Keep the geometry in mind; vectors are arrows and points, and a matrix is both a table of data and a transformation of space. Show your calculations.

Exercise 1 Let $\boldsymbol{u} = (2, -1)$ and $\boldsymbol{v} = (1, 3)$. Compute:

(a) $3\boldsymbol{u} - 2\boldsymbol{v}$;

(b) the length $\|\boldsymbol{u}\|$;

(c) the unit vector in the direction of $\boldsymbol{u}$.

Exercise 2 Express the vector $(5, 1)$ as a linear combination of $(1, 1)$ and $(1, -1)$; that is, find a and b with $a(1, 1) + b(1, -1) = (5, 1)$. What does it mean, geometrically, that this is always possible for any target vector in the plane?

Exercise 3 *Two stimuli, each encoded by the activity of three neurons (a population vector).*

Let $\boldsymbol{a} = (3, 4, 0)$ and $\boldsymbol{b} = (0, 4, 3)$.

(a) Compute the dot product $\boldsymbol{a} \cdot \boldsymbol{b}$ and the two lengths.

(b) Compute the cosine similarity $\cos\theta$, and the approximate angle.

(c) In one sentence, what does a high cosine similarity say about the two neural representations?

Exercise 4 Two stimuli are represented by the vectors $\boldsymbol{p} = (1, 2, 2)$ and $\boldsymbol{q} = (3, 2, 4)$. Compute the Euclidean distance $\|\boldsymbol{q} - \boldsymbol{p}\|$ between their representations.

Exercise 5 Let $\boldsymbol{a} = (2, 1, -1)$ and $\boldsymbol{b} = (1, -1, 1)$.

(a) Compute $\boldsymbol{a} \cdot \boldsymbol{b}$. Are the two vectors orthogonal?

(b) If two population vectors are orthogonal, what does that say about how the two stimuli are encoded?

Exercise 6 *Projection is the geometric heart of least-squares regression.*

Project $\boldsymbol{b} = (4, 2)$ onto $\boldsymbol{a} = (3, 0)$, using $\operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b} = \dfrac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}} \boldsymbol{a}$.

(a) Compute the projection.

(b) Compute the residual $\boldsymbol{b} - \operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b}$ and verify it is orthogonal to $\boldsymbol{a}$.

Exercise 7 *A linear readout maps a 2-feature input to a 2-neuron response via a weight matrix.*

Let $\boldsymbol{W} = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 3 \end{smallmatrix}\right)$ and the input $\boldsymbol{x} = (1, 2)$. Compute the response $\boldsymbol{W}\boldsymbol{x}$.

Exercise 8 Let $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ (a shear) and $\boldsymbol{B} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$ (a scaling).

(a) Compute $\boldsymbol{A}\boldsymbol{B}$.

(b) Compute $\boldsymbol{B}\boldsymbol{A}$.

(c) Are they equal? What does this illustrate about composing transformations?

Exercise 9 Let $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 1 \\ 2 & 4 \end{smallmatrix}\right)$.

(a) Compute $\det \boldsymbol{A}$.

(b) By what factor does $\boldsymbol{A}$ scale areas?

(c) Is $\boldsymbol{A}$ invertible? How do you know?

Exercise 10 Compute the inverse of $\boldsymbol{A} = \left(\begin{smallmatrix} 4 & 3 \\ 5 & 4 \end{smallmatrix}\right)$, and verify your answer by checking that $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$.

Exercise 11 *A simplified excitation–inhibition balance gives two linear equations.*

Solve the system $3x + 2y = 12, \qquad x + 2y = 8.$

Exercise 12 Solve the 3×3 system by elimination:

$$x + y + z = 6, \qquad 2x - y + z = 3, \qquad x + 2y - z = 2.$$

Exercise 13 Consider the three vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(1, 1, 0)$.

(a) Are they linearly independent? Justify.

(b) What is their rank, and what does that say about the space they span?

Exercise 14 *Dimensionality of a small neural population.*

Across many trials, the activity vectors of three neurons always satisfy exactly $\boldsymbol{n}_3 = 2\,\boldsymbol{n}_1 - \boldsymbol{n}_2$.

(a) Are the three neurons' activities linearly independent?

(b) What is the effective dimensionality (rank) of the population activity?

(c) What does this redundancy suggest about the neural code?

Exercise 15 Let $\boldsymbol{A} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$.

(a) Compute $\boldsymbol{A}(1, 0)$ and $\boldsymbol{A}(0, 1)$. What transformation does $\boldsymbol{A}$ perform?

(b) Compute $\boldsymbol{A}^2$ and interpret it geometrically.